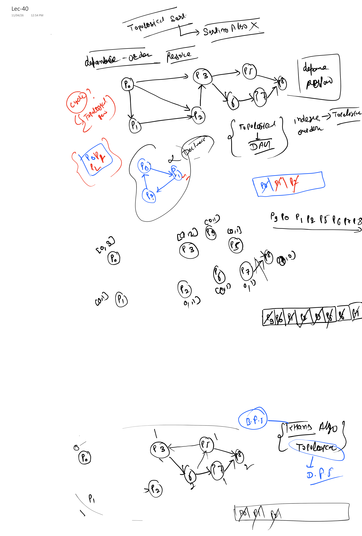
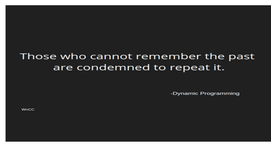
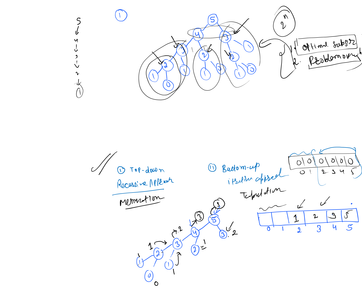
Lec-40

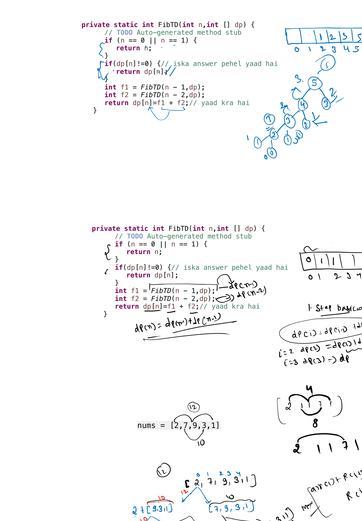






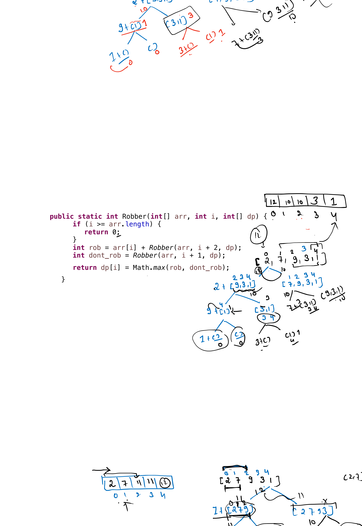

```
private static int FibTD(int n,int [] dp) {
    // TODO Auto-generated method stub
    if (n == 0 || n == 1) {
        return n;
    }
    if(dp[n]!=0) {// iska answer pehel yaad hai
        return dp[n];
    }
    int f1 = FibTD(n - 1,dp);
    int f2 = FibTD(n - 2,dp);
    return dp[n]=f1 + f2;// yaad kra hai
}
```

```
private static int FibTD(int n,int [] dp) {
    // TODO Auto-generated method stub
    if (n == 0 || n == 1) {
        return n;
    }
    if(dp[n]!=0) {// iska answer pehel yaad hai
        return dp[n];
    }
    int f1 = FibTD(n - 1,dp);
    int f2 = FibTD(n - 2,dp);
    return dp[n]=f1 + f2;// yaad kra hai
}
```

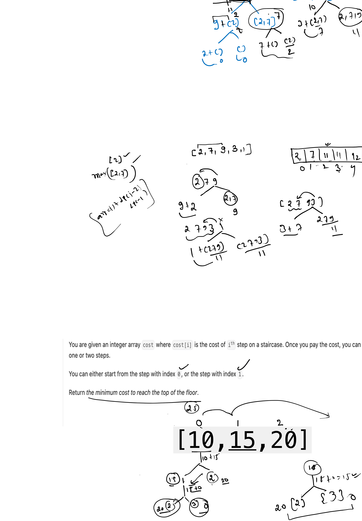



```
public static int Robber(int[] arr, int i, int[] dp) {
    if (i >= arr.length) {
        return 0;
    }
    int rob = arr[i] + Robber(arr, i + 2, dp);
    int dont_rob = Robber(arr, i + 1, dp);

    return dp[i] = Math.max(rob, dont_rob);
}
```



You are given an integer array `cost` where `cost[i]` is the cost of i^{th} step on a staircase. Once you pay the cost, you can either climb one or two steps.

You can either start from the step with index `0`, or the step with index `1`.

Return *the minimum cost to reach the top of the floor*.